Pipeline Automation with Airflow

In today's rapidly evolving data landscape, the ability to remotely manage and automate workflows is crucial for maintaining operational efficiency and scalability.

In this lesson, we'll delve into the capabilities of Apache Airflow's built-in REST API, an essential tool for enhancing interaction with the Airflow platform beyond the conventional web server UI.



What You'll Learn:

Airflow's REST API

> Understand the extensive functionality of Airflow's REST API and its applications for external system interactions.

Authentication methods for secure API access

> Identify and implement the correct authentication methods for secure API access, particularly in environments running Airflow on Docker.

Authentication methods with the Airflow API

Become adept at using basic authentication methods to secure communications with the Airflow API.

Managing Airflow using bash scripts ✖

Learn to use bash scripts effectively to list DAGs and manage various aspects of the Airflow instance.

Python for Airflow API interaction ✖

Develop the ability to utilise Python for advanced interactions with the Airflow API, enabling capabilities like triggering workflows and monitoring their execution remotely.

Ready to dive in?

Reflect:
In this lesson, we will explore how Airflow's REST API extends the functionality of workflow orchestration beyond the web UI. Think about how incorporating these capabilities could streamline your own data engineering pipelines. How might secure API interactions or Bash scripting help you automate DAG management at scale? Consider the ways you can integrate what you will learn in this lesson into a real-world project or use case.